Exercise Sheet 13

Throughout, the Clifford algebra $Cl(n)$ uses the convention $v^2 = -|v|^2$ for $v \in \mathbb{R}^n$. The group $\mathrm{Spin}(n)$ is the group of products of an even number of unit vectors, and $\rho(x)v = xvx^{-1}$. In $\mathbb{H} = Cl^0(3)$ we use $\mathbf{i} = e_2e_3$, $\mathbf{j} = e_3e_1$, $\mathbf{k} = e_1e_2$, and

$$\Phi(a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) = \begin{pmatrix} a + bi & -c - di \\ c - di & a - bi \end{pmatrix}.$$

For $g \in \mathrm{SU}(2)$, Π_m acts on homogeneous polynomials of degree m in z_1, z_2 by $(\Pi_m(g)f)(z) = f(g^{-1}z)$.

Exercise 13.1 (The group $\mathrm{Spin}(2)$). Let $J = e_1e_2 \in Cl(2)$.

(a) Show that $J^2 = -1$, that $Cl^0(2) = \{a + bJ : a, b \in \mathbb{R}\}$, and that $a + bJ \mapsto a + bi$ is an algebra isomorphism onto $\mathbb{C}$.

(b) Show that $\mathrm{Spin}(2) = \{\cos t + \sin t\, J : t \in \mathbb{R}\}$.

(c) Compute the matrix of $\rho(\cos t + \sin t\, J)$ in the basis e_1, e_2.

(d) Identify $\mathrm{Spin}(2)$ with $\mathrm{U}(1)$ by (a), and $\mathrm{SO}(2)$ with $\mathrm{U}(1)$ by sending the rotation through φ to $e^{i\varphi}$. Show that ρ becomes $z \mapsto z^2$. Find $\ker\rho$ and lift the loop of rotations through $\varphi \in [0, 2\pi]$.

Exercise 13.2 (Rotations from quaternions). Let $\omega = e_1e_2e_3 \in Cl(3)$ and $\iota(v) = -v\omega$ for $v \in \mathbb{R}^3$. Let $q = \frac{1}{2}(1 + \mathbf{i} + \mathbf{j} + \mathbf{k})$.

(a) Show that ω is central, that $\iota(e_1) = \mathbf{i}$, $\iota(e_2) = \mathbf{j}$, $\iota(e_3) = \mathbf{k}$, and that $\iota(\rho(x)v) = x\,\iota(v)\,x^{-1}$ for $x \in \mathrm{Spin}(3)$.

(b) Show that $q \in \mathrm{Spin}(3)$ and $q\mathbf{i} = \mathbf{j}q$. Explain why $q\mathbf{j}q^{-1} = \mathbf{k}$ and $q\mathbf{k}q^{-1} = \mathbf{i}$. Write the matrix of $\rho(q)$ in the basis e_1, e_2, e_3 and identify the rotation.

(c) Find the orders of q and $\rho(q)$. Show that $\Phi(q)$ has order 6 while its class in $\mathrm{PU}(\mathbb{C}^2)$ has order 3.

Exercise 13.3 (Optional extension: full turns and one circle subgroup). Let $u(\theta) = \mathrm{diag}(e^{i\theta/2}, e^{-i\theta/2}) \in \mathrm{SU}(2)$. Let $R(\theta) = \rho(u(\theta))$, the rotation through θ in the (e_2, e_3)-plane (the full-turn example, Example 13.9). Use the basis $f_k = z_1^{m-k}z_2^k$ of V_m, and let $\bar{\pi}_1$ be the projective representation of $\mathrm{SO}(3)$ defined in Corollary 13.18(c).

(a) Write the matrices of $\Pi_1(u(2\pi))$, $\Pi_1(u(4\pi))$ and $\Pi_2(u(2\pi))$. Which of them fix every ray?

(b) Show that $W(R(\theta)) = e^{i\theta/2}\Pi_1(u(\theta))$ is a well-defined continuous homomorphism on the circle group $C = \{R(\theta) : \theta \in \mathbb{R}\}$, and that $[W(R)] = \bar{\pi}_1(R)$ for $R \in C$.

(c) Why does (b) not contradict the nontrivial-multiplier statement of Corollary 13.18(c)?

Exercise 13.4 (Differentials on $\mathrm{O}(2)$). Let $r_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and $s = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ in $\mathrm{O}(2)$. Let $X = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, which spans $\mathfrak{so}(2)$.

(a) Show that $sr_\theta s^{-1} = r_{-\theta}$, that every element of $\mathrm{O}(2)$ is r_θ or $r_\theta s$, and that $\mathrm{O}(2)$ has two components.

(b) Show that the trivial character 1 and $\det$ have the same differential, but are not isomorphic.

(c) Which hypothesis of the integration results of Lecture 4 fails here? What data does $d\Pi$ miss for a representation Π of $\mathrm{O}(2)$?

Exercise 13.5 (The Heisenberg extension of $\mathbb{R}^2$). On $\mathbb{R}^2$ let $\omega(x,y) = x_1y_2 - x_2y_1$ and $c(x,y) = e^{i\omega(x,y)/2}$. On $E = \mathbb{R}^2 \times \mathbb{T}$ define

$$(x,\lambda)(y,\mu) = (x+y, \lambda\mu\, c(x,y)).$$

(a) Show that c is a multiplier.

(b) Show that E is a group with identity $(0,1)$ and central subgroup $\{0\} \times \mathbb{T}$, and find inverses.

(c) Compute the commutator relation between $(x,1)$ and $(y,1)$. Show that the projection $E \to \mathbb{R}^2$ has no homomorphic section $x \mapsto (x, b(x))$.

(d) Explain why passing to a universal cover does not help, and why this is consistent with the lifting theorem, Theorem 13.17.